

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and implemented **translational medicine strategies** across **preclinical and clinical development programs** by designing and executing **biomarker discovery and validation workflows** that reduced drug development cycle time by **18%** and informed critical go/no-go decision points for therapeutic asset advancement.
- Designed and executed **experimental programs** in collaboration with internal discovery, pharmacology, and clinical development teams and external **Contract Research Organizations (CROs)**, ensuring high-quality study execution, regulatory compliance, and **IND-ready documentation** for regulatory submissions.
- Interpreted, synthesized, and communicated **preclinical and clinical biomarker study results** using **statistical analysis, data modelling, and Power BI dashboards** to support multifunctional decision-making and translated complex datasets into actionable scientific insights for senior management and regulatory authorities.
- Prepared and reviewed **regulatory documentation including protocols, Investigator's Brochures, briefing meeting packages, and submission dossiers**, ensuring scientific rigor, **regulatory expectations**, and **GCP compliance** across **IND submissions** and pre-IND meeting deliverables.
- Managed external partners and vendor relationships through **Quality Management & Assurance protocols**, coordinated **change-control management** across interconnected clinical trial systems, and led cross-functional UAT and post-implementation triage to ensure **asset readiness for clinical development phase**.
- Initiated and implemented **cutting-edge bioanalytical technologies** and **assay development platforms** that demonstrated **scientific meaningfulness and reliability** for incorporation into both preclinical and clinical development programs, advancing mechanism of action understanding and biomarker measures for animal and patient studies.

SKILLS

Translational Medicine & Biomarker Science – biomarker discovery, biomarker validation, assay development, mechanism of action, drug response assessment, bioanalytical technologies.

Clinical Development & Regulatory Strategy – IND documentation, pre-IND meetings, regulatory submissions, Investigator's Brochures, GCP compliance, regulatory expectations, asset readiness.

Data Science & Analytics – Python, pandas, NumPy, scikit-learn, statistical analysis, hypothesis testing, data modelling, Power BI, Tableau, DAX, exploratory data analysis.

Study Design & Experimental Programming – preclinical study design, clinical trial support, biomarker study design, experimental program execution, investigative translational strategies.

Project & Partner Management – Contract Research Organization (CRO) management, multifunctional team coordination, change-control management, Quality Management & Assurance, vendor qualification, external stakeholder collaboration.

Technical & Database Systems – SQL, PostgreSQL, MySQL, Oracle, data warehousing, ETL pipelines, REST APIs, clinical trial data systems.

Domain Knowledge – transfusion medicine, plasma-derived therapeutics, drug discovery, preclinical development, clinical pharmacology, biomarker-driven development, healthcare regulatory compliance.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Translational Medicine Biomarker Discovery & Validation Program (HCL Technologies)

- Architected end-to-end **biomarker discovery and validation workflow** spanning preclinical animal models and clinical patient cohorts, integrating **statistical analysis in Python and R** with **Power BI dashboards** to track biomarker correlation to drug response and mechanism of action across **multiple therapeutic programs**.
- Designed **assay development protocols** and **bioanalytical method validation studies** with external **Contract Research Organizations**, coordinated **Quality Management reviews**, and prepared **IND-supporting assay documentation** that achieved regulatory approval and reduced assay qualification time by **22%**.
- Synthesized preclinical pharmacology and clinical efficacy data using **exploratory data analysis** and **statistical hypothesis testing** to identify and validate **biomarker measures for animal and patient studies**, enabling three therapeutic assets to advance to clinical development phase with **regulatory confidence**.

Clinical Trial Data Integration & Regulatory Documentation Framework (HCL Technologies)

- Built integrated **clinical trial data warehouse** leveraging **SQL, PostgreSQL, and Oracle** to consolidate preclinical study results, biomarker analytics, and safety/efficacy datasets, enabling **multifunctional team decision-making** and **regulatory submission readiness**.
- Developed **IND submission package automation** using **Python ETL pipelines** to extract, transform, and validate study-level data for **Investigator's Brochures, protocols, and briefing meeting packages**, reducing **regulatory documentation preparation time by 28%** and ensuring **GCP and regulatory compliance**.
- Collaborated with discovery, clinical development, and regulatory teams to translate **preclinical and clinical biomarker study results** into **regulatory strategies** and **investigative translational strategies**, directly supporting **pre-IND meetings** and advancing asset readiness for clinical development phase.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)